

Solution Requirements

Date	27 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system shall allow users (meteorologists/admins) to securely log in using username and password with optional OTP verification.	High
2	Authorization levels	The system shall support role-based access: Admin (full access), Analyst (data processing & prediction), Viewer (view-only access to results).	High
3	External interfaces	The system shall integrate with weather APIs (e.g., rainfall, humidity, river levels) and external datasets for real-time data collection.	High

4	Transactions processing	The system shall process incoming weather data, perform preprocessing, and send it to the ML model for prediction without data loss.	High
5	Reporting	The system shall generate dashboards and reports showing flood risk levels, prediction history, and visual analytics (charts/graphs).	Medium
6	Business rules, etc.	The system shall apply predefined thresholds (e.g., rainfall > X mm, water level > Y) and ML model logic (XGBoost) to classify flood risk	High
7	Compliance to laws or regulations	The system shall comply with data privacy standards and ensure secure handling of environmental and user data.	Medium
8	Other	The system shall allow exporting prediction results in formats like CSV/PDF and support alerts/notifications for high-risk conditions.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The system shall process input data and generate predictions quickly.	Response time < 2 seconds
2	Scalability	The system shall handle increasing data volume and multiple users.	Supports up to 10,000 concurrent users
3	Security & Data Privacy	The system shall ensure secure data transmission and storage using encryption techniques.	End-to-end AES-256 encryption ,secure login
4	Reliability & Availability	The system shall be highly available with minimal downtime	99.9% uptime per month
5	Usability & Accessibility	The system shall provide a user-friendly interface with easy navigation and accessibility compliance	Meets WCAG 2.1 standards
6	Other	The system shall be easy to maintain and deploy across different environments (cloud/local).	Modular design, supports cross-platform deployment